LEARNING ROAD-NETWORK RESILIENCE UNDER DISRUPTIONS WITH SPECTRAL GRAPH THEORY AND GRAPH NEURAL NETWORKS

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Abstract. We study the estimation of structural connectivity loss in spatial graphs under simultaneous edge disruptions. The target is the relative decrease of algebraic connectivity, the second-smallest eigenvalue of the weighted graph Laplacian. We compare an interpretable first-order approximation derived from the Fiedler vector with a compact scenario-aware graph convolutional network (ScenarioGCN). A leakage-resistant evaluation is performed on 10,000 disruption scenarios over random geometric graphs: scenarios derived from the same base graph are kept in a single train, validation, or test split. Across five independent seeds, the two methods obtain nearly identical mean absolute error (ScenarioGCN: 0.0656 ± 0.0132; spectral approximation: 0.0652 ± 0.0243). ScenarioGCN substantially improves root mean squared error and R^2, whereas the spectral approximation provides markedly better rank correlation. These results suggest complementary roles: learning controls large nonlinear errors, while spectral sensitivity remains a reliable, interpretable ranking tool. The present study is a synthetic benchmark and does not yet constitute a real flood-risk model.

1. Introduction

Road-network disruptions caused by extreme weather, construction, or local failures can alter global connectivity even when only a small subset of links is unavailable. A common computational task is therefore to rank many possible disruption scenarios. Exact spectral recomputation is well defined, but a fast surrogate may become useful when the number of networks and scenarios is large.

This work connects two viewpoints. Spectral graph theory supplies a precise connectivity measure and an interpretable local sensitivity. Graph neural networks (GNNs) provide a data-driven mechanism for representing interactions among several failed edges. Instead of presuming that the learned model must dominate the mathematical approximation, we ask where each method is reliable.

Our preliminary contributions are:

(1) a normalized prediction target comparable across graphs;
(2) an explicit first-order Fiedler-vector baseline;
(3) a small graph-level regressor requiring only standard PyTorch;
(4) a graph-disjoint experimental split that prevents structural leakage;
(5) a five-seed comparison stratified by disruption size.

2. Mathematical formulation

Let $G = (V, E, w)$ be a connected, undirected graph with positive edge weights. Its weighted adjacency matrix is A, degree matrix is D, and combinatorial Laplacian is $L = D - A$. Write the ordered eigenvalues as

$$0 = \lambda_1(L) \leq \lambda_2(L) \leq \cdots \leq \lambda_{|V|}(L).$$

Fiedler [1] introduced λ_2 as the algebraic connectivity. It is positive exactly when an undirected graph is connected.

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Definition 2.1 (Relative connectivity loss). *For a set of disrupted edges $S \subseteq E$, let $G - S$ denote the damaged graph. We define*

$$y(G,S) = \mathrm{clip}\left(1 - \frac{\lambda_2(G-S)}{\lambda_2(G)}, 0, 1\right). \tag{1}$$

Thus $y = 0$ represents no measured loss, while $y = 1$ includes disconnection.

Suppose λ_2 is simple and u_2 is its unit eigenvector. For edge $e = (i,j)$, standard symmetric eigenvalue perturbation gives

$$\frac{\partial \lambda_2}{\partial w_e} = (u_{2,i} - u_{2,j})^2. \tag{2}$$

Removing several edges is a finite, not infinitesimal, perturbation. We use the clipped first-order prediction

$$\widehat{y}_{\mathrm{spec}}(G,S) = \mathrm{clip}\left(\frac{\sum_{(i,j)\in S} w_{ij}(u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1\right). \tag{3}$$

This expression ignores higher-order interaction and can be inaccurate near an eigenvalue crossing or disconnection. Its advantages are transparency and the absence of learned parameters.

3. ScenarioGCN

For each scenario, the model receives the normalized adjacency matrix of $G - S$, augmented with self-loops, and five node features: normalized degree in G, fraction of incident edges in S, absolute normalized Fiedler coordinate, and two spatial coordinates. Three graph convolution layers use the update

$$H^{(\ell+1)} = \mathrm{ReLU}\left(\widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}H^{(\ell)}W^{(\ell)}\right),$$

following the localized propagation principle of Kipf and Welling [2]. Mean and max graph pooling are concatenated with the mean failed-edge incidence rate. A two-layer head with sigmoid output predicts Equation (1). The model has three 48-dimensional hidden layers, dropout 0.15, and is trained using AdamW and mean squared error.

4. Experimental design

4.1. **Synthetic spatial graphs.** Random geometric graphs place nodes in the unit square and connect pairs within a radius. They are a basic spatial-network model rather than a faithful road generator. We adapt the radius until a connected graph with 35–65 nodes is obtained. Edge conductance is set to inverse Euclidean length. For each base graph we sample 20 scenarios, removing between one and eight edges.

The full benchmark contains five independent runs of 2,000 scenarios. Within each run, 70%, 15%, and 15% of the scenarios are assigned to training, validation, and test sets by *base graph identifier*. Consequently no damaged version of a training graph appears in the test set. Each run has 1,400 training, 300 validation, and 300 test scenarios, for a pooled total of 1,500 held-out predictions.

4.2. **Evaluation.** We report mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and Spearman rank correlation ρ. The constant baseline predicts the training-target mean. Mean and sample standard deviation are computed over the five seeds (11, 22, 33, 44, 55), treating the seed rather than an individual correlated scenario as the independent replicate.

TABLE 1. Held-out performance, mean ± standard deviation over five seeds. Lower is better for MAE/RMSE; higher is better for R^2 and ρ.

Method	MAE	RMSE	R^2	Spearman ρ
ScenarioGCN	0.0656 ± 0.0132	$\mathbf{0.1492 \pm 0.0273}$	$\mathbf{0.5464 \pm 0.0952}$	0.6279 ± 0.0571
Spectral first order	$\mathbf{0.0652 \pm 0.0243}$	0.2090 ± 0.0508	0.1331 ± 0.1222	$\mathbf{0.9507 \pm 0.0172}$
Training-mean constant	0.1200 ± 0.0217	0.2244 ± 0.0423	-0.0077 ± 0.0119	0

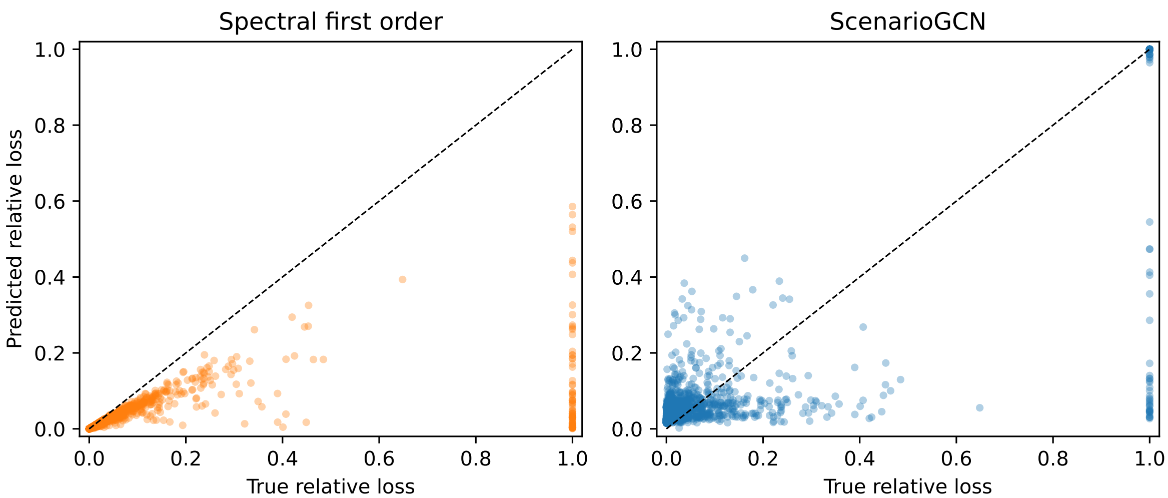

FIGURE 1. Pooled held-out predictions from five independent seeds. The dashed line is exact agreement. Clipping in the first-order baseline creates a band at prediction one; ScenarioGCN is more conservative at extremes.

5. RESULTS

Table 1 shows that MAE is effectively tied: the difference of 0.0004 is negligible relative to between-seed variation. ScenarioGCN reduces mean RMSE by approximately 28.6% relative to the spectral approximation and achieves positive R^2 in every seed. In contrast, spectral sensitivity has much higher and more stable rank correlation. It therefore orders scenarios well but occasionally makes large finite-perturbation errors; the learned model controls these errors while sacrificing ranking resolution.

Figure 2 indicates that performance depends on disruption size. The first-order method is especially competitive for small perturbations, as expected from Equation (2). ScenarioGCN becomes competitive for moderate and larger perturbations, although variability across generated graphs remains substantial. These observations support a hybrid research direction rather than unconditional replacement of the spectral model.

6. DISCUSSION

The experiment answers a narrower question than real road resilience. It measures structural connectivity, not travel demand, capacity, delay, or recovery time. Moreover, random edge deletion is not a flood model. Flooded links are spatially correlated and depend on elevation, drainage, rainfall, and road geometry. Accordingly, the present results must be described as “disruption scenarios”, not flood predictions.

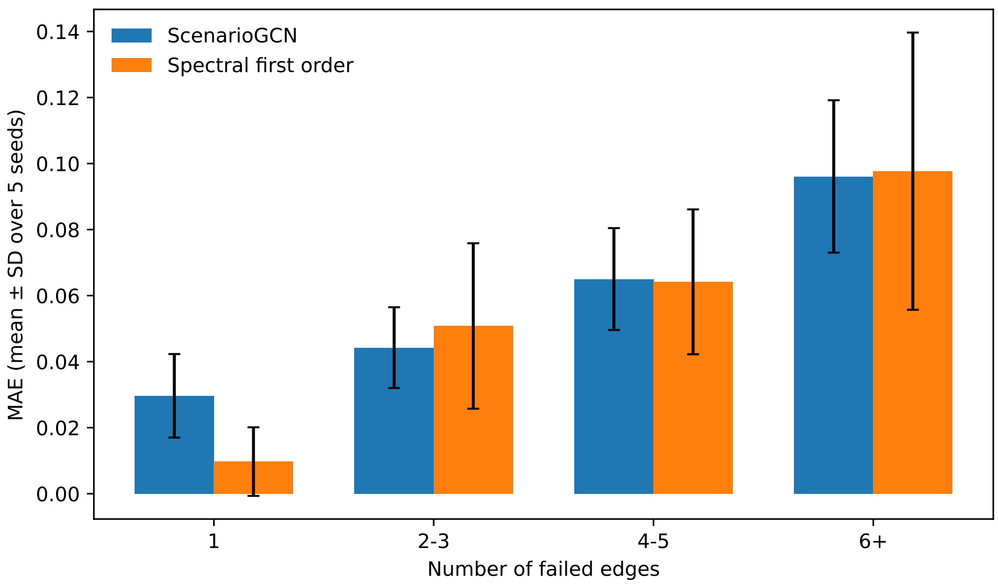

FIGURE 2. MAE stratified by disruption size. Error bars show standard deviation over seeds, not scenario-level uncertainty.

The evaluation also exposes a useful methodological lesson. Randomly splitting all scenarios would allow damaged variants of one base graph in both training and test sets, producing optimistic estimates. Graph-level splitting should be mandatory in this setting. Finally, a GCN uses local message passing and may struggle to reconstruct a global eigenvalue; the strong R^2 but moderate rank correlation motivates residual learning, where the network predicts the error of Equation (3) rather than the full target.

7. Limitations and next experiments

Before presenting final claims, the following preregistered extensions are required:

(1) repeat the experiment without the Fiedler coordinate to quantify how much learned performance is inherited from spectral information;
(2) add spatially clustered failures and test out-of-range graph sizes;
(3) compare direct GCN regression with a residual spectral–GCN hybrid;
(4) measure inference time under a fixed batching and hardware protocol;
(5) validate on a small, versioned OpenStreetMap street network acquired using OSMnx [3];
(6) only after obtaining a licensed hazard layer, study flood-conditioned failure probabilities.

No hypothesis test is emphasized with only five independent seeds. The current effect sizes and standard deviations are more informative than treating 1,500 dependent test scenarios as independent replicates.

8. Conclusion

On a leakage-resistant synthetic benchmark, ScenarioGCN and first-order spectral sensitivity have nearly equal average absolute error but distinctly different strengths. ScenarioGCN improves squared-error performance, while the Fiedler baseline preserves excellent scenario ranking. This complementary behavior provides a defensible mathematical–AI poster result and suggests a hybrid residual model as the next step. Claims about urban flooding or policy should wait for real-network and hazard-data validation.

Reproducibility statement

The complete data generator, spectral baseline, ScenarioGCN implementation, graph-disjoint split, five benchmark seeds, aggregation script, and generated tables are provided with the project. The reported runs used Python 3.13.11, standard CPU PyTorch, 2,000 scenarios per seed, at most 100 epochs, and fixed seeds 11, 22, 33, 44, and 55.

References

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